FEDERAL SPECIFICATION

LINER MATERIAL, GREASEPROOF, (FOR SPIRALLY-WOUND FIBER CONTAINERS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for greaseproof liner material intended for use as the first inner ply of spirally-wound fiber containers.

1.2 Classification.

1.2.1 *Types.* Greaseproof liner material shall be of the following types, as specified.

Type I.—Kraft-film laminate (see 6.6).
Type II.—Vegetable parchment (see 6.7).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

UU-P-31—Paper, General Specifications and Methods of Testing.
UU-P-271—Paper, Wrapping, Waterproofed, Kraft.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-D-723—Drums, Fiber.
PPP-T-76—Tape, Paper, Water-Resistant.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-B-121—Barrier Material, Greaseproofed, Flexible (Waterproofed).
MIL-G-3278—Grease, Aircraft and Instrument (For Low and High Temperature).
MIL-A-140—Adhesive, Water-Resistant, Waterproof Barrier-Material.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

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MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Greaseproof liner material shall be made from such materials and by such processes as to insure compliance with this specification.

3.2 Construction. Greaseproof liner material shall be constructed of one or more plies or components in such a manner as to insure compliance with the requirements of this specification. Coatings or laminations shall be uniform.

3.3 Form. Greaseproof liner material shall be furnished in rolls of such length and width as to be suitable for use in the manufacture of fiber cans.

3.4 Color identification. Greaseproof liner material shall be marked so as to be readily visible from the greaseproof side by means of distinctly red dots approximately 3/8″ to 1/2″ in diameter, placed so as to form lines parallel to the machine direction of the material and to each other, and spaced 6″ ±1/2″ apart. The dots within a line should be spaced approximately 1/4″ to 3/8″ apart so as to readily differentiate this material from grade A of Military Specification MIL–B–121.

3.5 Certification. The manufacturer of the liner material shall be required to submit a certificate indicating compliance with all requirements of this specification at the time of invitation for bids. Acceptance of the material by the Government may be on the basis of this certificate, or by lot acceptance testing at the manufacturer's plant at the discretion of the Government inspector. If testing facilities at the manufacturer's plant are not acceptable to the Government inspector, a testing facility which is satisfactory shall be engaged.

3.6 Greaseproofness. The *uncreased* greaseproof liner material shall resist the penetration of grease for a minimum of 24 hours and shall show no evidence of delamination, embrittlement, or disintegration when tested as specified in 4.3.1.

3.7 Physical properties. Greaseproof liner material shall possess the minimum average physical strength values given in table I when tested as specified in 4.3.2:

TABLE I.—*Minimum average physical characteristic*

	Type I	Type II
Bursting strength, points	30	20
Tearing resistance, grams, weakest principal direction	100	25
Tensile strength, lbs/inch width, weakest principal direction ..	20	15

3.8 Chemical properties. The liner material shall have a pH range of 6.5 to 7.5, with a maximum acid or alkaline content of 0.02 percent (as SO_3 or NaOH), when tested as specified in 4.3.3.

3.9 Roll identification. Each roll of liner material shall be marked as follows:

Title, date and number of this specification.
Manufacturer's name and address.
Manufacturer's designation (trade name), if any.
Month and year of manufacture.

3.10 Workmanship. Greaseproof liner material shall be uniformly constructed, free from holes, tears, cuts, sharp creases, wrinkles, or other defects which might impair the usefulness of the material for its intended purposes. Finished material shall not block to such an extent that will cause tearing, delamination, or other injury to the material when unrolled.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. A lot shall consist of all material produced in one unchanged process by one manufacturer and submitted for inspection at one time.

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4.2 Sampling.

4.2.1 *Lot acceptance.* At the discretion of the Government inspector, samples selected in accordance with level L–7 of Military Standard MIL–STD–105 for normal inspection and level L–5 for reduced inspection shall be subjected to the tests specified in 4.3. In applying these levels to rolls, the inspection lot size shall be expressed in units of square yards of material, and the sample size of square yards shall be apportioned among the rolls so that approximately 5 yards are inspected from each roll.

4.2.2 *Visual and dimensional inspection.* Samples selected in accordance with level II of Military Standard MIL–STD–105 shall be inspected for defects as set forth below. Inspection for major defects shall be at an AQL of 0.65 percent defective and for minor defects at an AQL of 1.5 percent defective.

4.2.2.1 Defects in appearance, construction, and general workmanship. (Check both sides of the material).

Examine	Defects
Cleanliness (major)	More than usual amount of dirt resulting from transportation and handling. Presence of grease, dirt, etc., sufficient to render material unfit for intended use.
Workmanship (major)	Any tears, cuts, holes, sharp creases, wrinkles, delamination, embrittlement, cracks, blisters, abrasions, blocking, excessive curl, selvage, etc.
Color identification (major)	Not visible from greaseproof side. Stripes differ from that specified in 3.4. Dimensions and spacing of stripes differ from that specified in 3.4.
	Missing.
Roll identification (major)	Differs from that specified in 3.9.
	Missing.
Construction (major)	Not uniform.
Coating (major)	Not uniform, (when applicable).
Width (minor)	Varies by ± ⅛ inch from size specified in contract.
Unwinding of roll (minor)	Not suitably restrained from unwinding. Not wound evenly and tightly, causing wrinkles, creases, or telescoping.

4.3 Tests.

4.3.1 *Greaseproofness.*

4.3.1.1 *Preparation of reagent.* To 100 milliliters of pure gum spirits of turpentine, U.S.P. Grade, (specific gravity 0.860 to 0.875 at 60°F.), add 5 grams of anhydrous calcium chloride and 1 gram of an oil soluble red dye, Calco Oil Red N–1700 or approved substantial equal (see 6.3). Stopper the container, shake well, and let stand at least 10 hours. Filter through dry filter paper and retain the filtrate in a dry bottle. To each 100 grams of the filtrate add 25 grams of grease, Beacon Grease 325 or approved substantial equal (see 6.3) conforming to Military Specification MIL–G–3278. Agitate the turpentine grease mixture by means of a Waring Blender or substantial equal (see 6.3) until the grease is thoroughly dissolved. Store the reagent in a dry bottle having a tightly fitting glass stopper.

4.3.1.2 *Procedure.* Cut 20 specimens each 4 inches square and condition for not less than 24 hours at 73° ±2°F. and 50 ±5 percent relative humidity. Place each specimen (*uncreased*), greaseproof side up (if applicable), on a piece of white Whatman number 2 filter paper (or approved equal) (see 6.3) that will extend at least ½ inch beyond all edges of the specimen. Place a metal ring, weighing ¼ to ½ pound and having dimensions of 5¾ ±1/16 inches outside diameter and 3¾ ±1/16 inches inside diameter on the specimen so that all corners and edges are held in contact with the filter paper. Place 5 grams of sand, screened to pass a number 20 U. S. Standard Sieve and to be retained on a number 30 U. S. Standard Sieve, in the center of the *uncreased* specimen and add, by means of a pipette, 1.5 milliliters of the prepared test reagent to the sand. On each specimen place a 50 to 55 gram weight, centered over the saturated sand, to hold the specimen in contact with the filter paper. At the end of 2 hours, and every hour thereafter for 6 hours, add 0.5 milliliters of test reagent to the sand to replace that lost by evaporation. Conduct this test at 73° ±2°F. and 50 ±5 percent relative humidity. At the

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end of 24 hours, lift the specimen and examine the filter paper at the point of contact. Any evidence of staining or grease spots on the filter paper, or of delamination, embrittlement, or disintegration of the specimen shall constitute failure to pass the test.

4.3.1.3 *Acceptance.* The material shall be considered acceptable if no more than one failure occurs in the original 20 specimens. If two specimens in the original 20 show penetration by the grease reagent, or evidence of delamination, embrittlement, or disintegration, the material shall be retested. (Mere staining of the back of the specimen by the grease reagent or dye without transfer to the filter paper shall not constitute a failure). If retesting is done, 20 specimens shall be used. If no more than one failure occurs in the retest, the material shall be considered acceptable. Three or more failures occurring in the original 20 specimens tested will be sufficient cause for rejection of the material without retesting.

4.3.2 *Physical properties.* Tests for physical strength properties shall be conducted in accordance with the following methods as described in Federal Specification UU–P–31:

Test	Method
Bursting strength	112
Tearing resistance	170
Tensile strength	171
Atmospheric conditions	102

4.3.3 *Chemical Properties.*

4.3.3.1 *Hydrogen ion concentration (pH).* This determination shall be conducted in accordance with method 200 of Federal Specification UU–P–31, with the following changes:

Increase individual sample size to 5 grams.
Increase size of Erlenmeyer flasks to 500 ml.
Increase total amount of distilled water used per sample to 250 ml.
Conduct test in triplicate.
Wear rubber gloves when cutting samples to avoid contamination.

If the average of the three tests does not fall within the range specified for the material, the lot represented by the samples shall be rejected. Conduct a blank determination as described above using 250 ml. of distilled water without the sample.

4.3.3.2 *Acid or alkaline content.* Using the three sample extracts and blank on which the pH determination was made as described (see 4.3.3.1) above, titrate these solutions to the pH value of the blank, using 0.01 *N* HCl or 0.01 *N* NaOH solutions, whichever is appropriate. Calculate any acidity (in terms of SO_3) or alkalinity (in terms of NaOH), as follows:

$$\text{Percent } SO_3 = \frac{T \times N \times 4}{W}$$

where:

T = ml. NaOH required to titrate the extract to the pH of the blank.
N = Normality of NaOH solution.
W = Weight of test specimen.

or

$$\text{Percent NaOH} = \frac{T \times N \times 4}{W}$$

where:

T = ml. of HCl required to titrate the extract to the pH of the blank.
N = Normality of NC1 solution.
W = Weight of test specimen.

Three determinations shall be made and values averaged to determine compliance with the requirement specified in 3.8.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging. Unless otherwise specified by the procuring activity, liner material shall be packaged as follows:

5.1.1 *Levels A, B, and C.* Each roll shall be wound on a substantial core with an outside diameter of not less than 3 inches and of sufficient length so that it will project approximately

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½ inch beyond the edge of the roll at each end. The roll shall contain 100 yards or even multiples thereof (see 6.2) and shall be suitably restrained from unwinding. Each roll shall be completely wrapped and headed with at least one thickness of class C–1 or better, waterproof barrier material conforming to the requirements of Federal Specification UU–P–271. Waterproof barrier material shall be sealed with waterproof adhesive conforming to the requirements of Military Specification MIL–A–140 or with 3 inch wide tape conforming to the requirements of Federal Specification PPP–T–76, centered along the end edge of the waterproof barrier material and applied across the full width of the roll.

5.2 Packing.

5.2.1 *Level A.* Two rolls packaged as in 5.1.1 shall be overpacked in a nailed wood box, wood-cleated-plywood box, or wirebound box, constructed, sealed, and strapped in conformance with the requirements of Federal Specification PPP–B–621, class 2; PPP–B–601 (overseas type); or PPP–B–585, class 2 or 3 for type 2 load.

5.2.1.1 Alternatively each roll of liner material packaged as in 5.1.1 may be packed in a fiber drum conforming to the requirements of Federal Specification PPP–D–723, type III. No overpacking is required.

5.2.2 *Level B.* Rolls of liner material packaged as in 5.1.1 shall be packed in exterior containers conforming to the requirements of Federal Specification PPP–B–621, class 1; PPP–B–601 (domestic type); or PPP–B–585, class 1, as specified.

5.2.2.1 Alternatively each roll of liner material packaged as in 5.1.1 may be packed in a fiber drum conforming to the requirements of Federal Specification PPP–D–723, type I. No overpacking is required.

5.2.3 *Level C.* Unless otherwise specified rolls of liner material shall be packed in accordance with suppliers best commercial practice, so as to insure acceptance and safe delivery by common or other carriers for safe transportation at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order (see 6.2), individual rolls and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.3.2.1 *Precautionary marking.* The following marking shall appear on at least one side of each wrapped roll, drum, or box in letters not less than ¾ in. in height.

STAND ON END
KEEP COOL AND DRY

6. NOTES

6.1 Intended use. The greaseproof liner material covered by this specification is intended for use as the first inner ply of spirally wound fiber containers. This material is *NOT* to be used in applications requiring grade A, Military Specification MIL–B–121, greaseproof barrier material.

6.2 Ordering data. Procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Length and width (see 5.1.1).

(*c*) Selection of the applicable level of packaging, packing (see section 5).

(*d*) Special marking.

6.3 Since the use of other than specified reagents, etc., may affect the test results, substitute materials should be used only after it

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has been determined that their use will not adversely affect test results.

6.4 Calco Oil Red N–1700 may be obtained from the American Cyanamid Company, Calco Chemical Division, Bound Brook, New Jersey.

6.5 Beacon Grease 325 may be obtained from the Standard Oil Company of New Jersey, Elizabeth, New Jersey.

6.6 A lamination of a 25 pound per ream (24″ x 36″—500) glassine paper to a 30 pound per ream (24″ x 36″—500) kraft paper has been found to meet the requirements of this specification for type I material.

6.7 A 35 lb. basis weight (24″ x 36″—500) vegetable parchment paper has been found to meet the requirements of this specification for type II material.

6.8 Transportation description. Transportation description applicable to this item is:

Type I

Paper, wrapping, not otherwise indexed by name, oiled or waxed.
Carload minimum weight 36,000 pounds.
Motor volume minimum weight 36,000 pounds.

Type II

Paper, vegetable parchment lining.
Carload minimum weight 36,000 pounds.
Motor volume minimum weight 36,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, *or conveying any rights or permission to manufacture,* use, or sell any patented invention that may in any way be related thereto.

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